

Required space for placing a boat is too small

MC-101334 Collision

Status	Resolved / Fixed
Affected	Minecraft 1.9.2, Minecraft 16w14a, Minecraft 16w15a, Minecraft 16w15b, Minecraft 1.9.3 Pre-Release 1, Minecraft 1.9.3 Pre-Release 2 (+161 more)
Fixed in	1.18 Pre-release 1
Reported	2016-04-26

STEPS TO REPRODUCE

1. Launch Minecraft 18w31a.
2. Create a Creative world.
3. Put a boat in your hand.
4. Place a block on the ground.
5. Look at the ground directly in front of the placed block.
6. Hold the right-click button while slowly moving the cursor away from the block.
7. Observe that the boat is placed with its collision box intersecting the block, causing the boat to fall into the void.

OBSERVED BEHAVIOUR

Actual behavior

The boat can be placed when there is not enough unobstructed space for its full collision box. The placement check uses the boat's bounding box reduced by `0.1` blocks on each axis:

```
```java
entityboat.getEntityBoundingBox().grow(-0.1D)
```
```

As a result, the check may report no collision even though the boat's actual collision box intersects the block directly in front of the player. The boat is then spawned partially inside the block. In 18w31a, this can cause the boat to fall into the void instead of remaining properly positioned.

EXPECTED BEHAVIOUR

When placing a boat near a block, the game should require enough unobstructed space for the boat's full collision box. If any part of the boat would intersect the nearby block or other solid blocks, placement should fail rather than create the boat in an invalid position. Successfully placed boats should remain properly supported and not fall into the void due to intersecting terrain.

ENVIRONMENT

Minecraft Java Edition Versions: 1.9.2, 16w14a, 16w15a, 16w15b, 1.9.3 Pre-Release 1, 1.9.3 Pre-Release 2, 1.9.3 Pre-Release 3, 1.9.3, 1.9.4, 16w20a, 16w21a, 16w21b, 1.10 Pre-Release 1, 1.10 Pre-Release 2, 1.10, 1.10.1, 1.10.2, 16w32a, 16w32b, 16w33a, 16w35a, 16w36a, 16w38a, 16w39a, 16w39b, 16w39c, 16w40a, 16w41a, 16w42a, 16w43a, 16w44a, 1.11 Pre-Release 1, 1.11, 16w50a, 1.11.2, 17w06a, 17w13a, 17w13b, 17w14a, 17w15a, 17w16a, 17w16b, 17w17a, 17w17b, 17w18a, 17w18b, 1.12 Pre-Release 1, 1.12 Pre-Release 2, 1.12 Pre-Release 3, 1.12 Pre-Release 4, 1.12 Pre-Release 5, 1.12 Pre-Release 6, 1.12 Pre-Release 7, 1.12, 1.12.1 Pre-Release 1, 1.12.1, 1.12.2 Pre-Release 1, 1.12.2 Pre-Release 2, 1.12.2, 17w43a, 17w43b, 17w45a, 17w45b, 17w46a, 17w47a, 17w47b, 17w48a, 17w49a, 17w49b, 17w50a, 18w01a, 18w02a, 18w03b, 18w05a, 18w06a, 18w07a, 18w07b, 18w07c, 18w08a, 18w08b, 18w09a, 18w10a, 18w10b, 18w10c, 18w10d, 18w11a, 18w14a, 18w14b, 18w15a, 1.13-pre1, 1.13-pre6, 1.13, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1-pre2, 1.13.1, 1.13.2-pre1, 1.13.2-pre2, 1.13.2, 18w43a, 18w43b, 18w43c, 18w44a, 18w48a, 18w48b, 18w49a, 18w50a, 19w12b, 19w13b, 19w14a, 19w14b, 1.14.3, 1.14.4, 19w34a, 19w39a, 19w42a, 19w45b, 1.15 Pre-release 1, 1.15, 1.15.1, 1.15.2 Pre-release 2, 1.15.2, 20w06a, 20w07a, 20w08a, 20w09a, 20w10a, 20w11a, 20w12a, 20w13b, 20w18a, 1.16 Release Candidate 1, 1.16, 1.16.1, 1.16.2 Pre-release 1, 1.16.2 Release Candidate 1, 1.16.2 Release Candidate 2, 1.16.2, 1.16.3, 1.16.4 Pre-release 1, 1.16.4 Release Candidate 1, 1.16.4, 20w45a, 20w46a, 20w49a, 21w03a, 1.16.5, 21w05a,

21w05b, 21w06a, 21w07a, 21w11a, 21w13a, 21w14a, 21w15a, 21w16a, 1.17 Pre-release 2, 1.17 Pre-release 5, 1.17 Release Candidate 1, 1.17, 1.17.1 Release Candidate 1, 1.17.1, 21w37a, 21w40a, 21w43a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

The required space when placing a boat is too small.

This allows the collision box of the boat to intersect with blocks.

At 18w31a will make the boat fall into the void.

Video:

<https://www.youtube.com/watch?v=MIRPQVz-y9Y>

How to reproduce

- Put a boat in your hand
- Place a block on the ground
- Look at the ground right in front of the placed block
- Hold right click while slowly moving the cursor away from the block

Code analysis

Based on 1.12.2 decompiled using MCP 9.40

The method `net.minecraft.item.ItemBoat.onItemRightClick(World, EntityPlayer, EnumHand)` decreases the collision box size when it tests for no collisions:

```
if (!worldIn.getCollisionBoxes(entityboat, entityboat.getEntityBoundingBox().grow(-0.1D)).isEmpty())  
{  
    return new ActionResult<ItemStack>(EnumActionResult.FAIL, itemstack);  
}
```

MANUAL RESULT

| | |
|------------------|--------------------|
| Version used | |
| Reproduced? | YES / NO / PARTIAL |
| Crash file | |
| Notes | |
| | |
| Tested by / date | |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-101334>